SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

COURSE NAME: Theory of Computation **COURSE CODE:** CSA1301

COURSE OUTCOME COVERED

CO1: ***Apply mathematical foundations such as formal proofs, induction, and basic concepts of automata theory to solve computational problems (BL:3)***

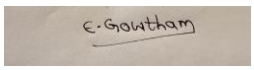
Assessment Tool Weightage: 10%

QUESTIONS: 5 **Total Marks:50**
Duration : 1 Hour

Technical Presentation

Code of Conduct:

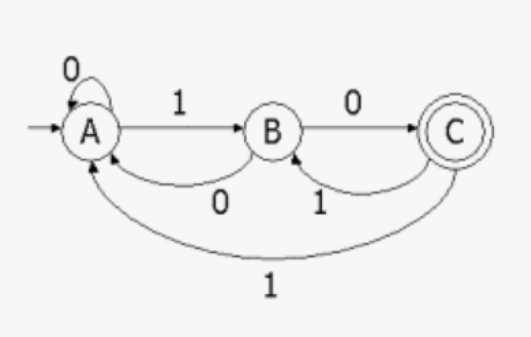
I Enamala Gowtham/192311190 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

1.How can every Non-Deterministic Finite Automaton (NFA) be converted into an equivalent Deterministic Finite Automaton (DFA), and what are the challenges involved? Consider the

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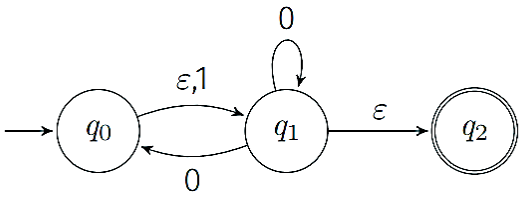
NFA given below which accepts some language L over the alphabet Σ={0,1}. Construct a DFA that represents the same language L



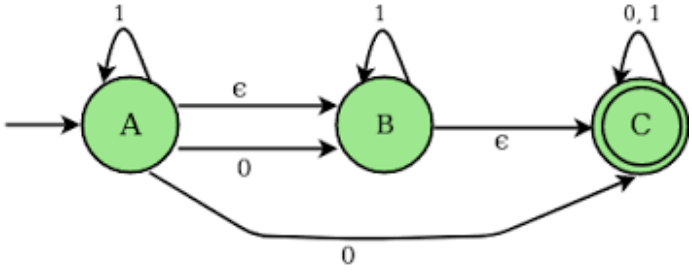
2.How is mathematical induction used to prove properties of strings and languages in automata theory? Apply the principle of mathematical induction and prove that for all positive integers n, the statement given below is true.

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

3.What role do epsilon (ε) transitions play in finite automata, and how can ε-NFA be converted into an equivalent NFA without epsilon transitions? construct an equivalent NFA without ε-transitions that accepts the same language. Draw its transition table and mark the initial and final states



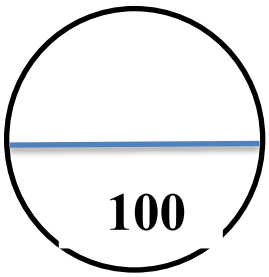
4.A NFA with ε-transitions is given below. Find ε-closure for each state and construct an equivalent DFA which accepts the same language. Draw its transition table and mark the initial and final states



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RUBRICS FOR CALCULATION

Criteria	Weightage	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	3	3	3	3	15
Accuracy of Answer	25	2.5	2.5	2.5	2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50



CO1 Assessment - 3

Name : E. Gowtham

Reg. No : 192311190

Course : Theory of computation

Code : CSA 1301

Conversion of NFA into DFA

A Non-Deterministic finite Automation (NFA) can be converted into an equivalent Deterministic finite Automation (DFA) using the subset construction method.

Steps

1. The initial DFA state is the set containing the initial NFA state
2. find the transitions for inputs 0 and 1.
3. Create a new DFA state for every distinct set of NFA states obtained.
4. Repeat until no new states are generated
5. Any DFA state containing an NFA final state becomes a final state

challenges: The number of DFA states can grow exponentially, up to (2^n) 2^n for an NFA with n states. Identifying all reachable subsets and avoiding duplicate states can also be difficult.

Given NFA

From the diagram in the question, the transiti

State	Input 0	Inp
A	A	B
B	A, C	∅
C	∅	A, B

Initial state: A

Final state: C

Equivalent DFA transition table

let the DFA states be:

$S_0 = \{A\}$

$S_1 = \{B\}$

$S_2 = \{A, C\}$

$S_3 = \{A, B\}$

$S_D = \emptyset$

DFA State	0	1
$S_0 = \{A\}$	S_0	S_1
$S_1 = \{B\}$	S_2	S_D
$S_2 = \{A, C\}$	S_0	S_3
$S_3 = \{A, B\}$	S_2	S_1
$S_D = \emptyset$	S_D	S_D

Initial state: S_0

final state: S_2, because it contains C

Result: The constructed DFA accepts the same language as the given NFA.

Mathematical induction in automata theory

Mathematical induction is a method used to prove that a statement is true for all positive integers. In automata theory, it is useful for proving properties of strings, languages, and automata.

Steps of mathematical induction

1. Base case: Prove that the statement is true for the smallest value, usually $n=1$
2. Inductive hypothesis: Assume that the statement is value true for $n=k$
3. Inductive step: using the assumption for $n=k$, prove that the statement is true for $n=k+1$.
4. Conclusion: Therefore, the statement is true for all positive integer n.

Example

Prove that

$$1+2+3+\cdots\cdots+n = \frac{n(n+1)}{2}$$

Step 1: Base case, $n=1$

$$1 = \frac{1(1+1)}{2} = 1.$$

Hence, the statement is true

Step2 : Inductive hypothesis

Assume that

$$1+2+\cdots+k = \frac{k(k+1)}{2}$$

Step3: Inductive step

for $n=k+1$.

$$1+2+\cdots+k+(k+1)$$

using inductive hypothesis,

$$= \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{(k+1)(k+2)}{2}$$

Hence, the statement is true for $k+1$

Conclusion: By mathematical induction,

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

is true for every positive integer n

Q3. Epsilon (ε) transitions and conversion of ε - NFA into NFA

An epsilon(ε) transitions allows an automation to move from one state to another without consuming any input symbol.

An ε-NFA can be converted into an equivalent NFA without ε-transitions by

calculating the ε-closure of each state and updati-ng the transitions and final states.

Given ε-NFA

The diagram contains states q_0, q_1, q_2, where q_0 is initial and q_2 is final

Transitions:

- $q_0 \xrightarrow{\varepsilon} q_1$
- $q_0 \xrightarrow{1} q_1$
- $q_1 \xrightarrow{0} q_1$
- $q_1 \xrightarrow{0} q_0$
- $q_1 \xrightarrow{\varepsilon} q_2$

Step-1; find ε closure

The ε-closure of a state includes the state itself and all states reachable using only ε-transition

State	ε-closure
q_0	$\{q_0, q_1, q_2\}$
q_1	$\{q_1, q_2\}$
q_2	$\{q_2\}$

Step 2: Construct the equivalent NFA

State	0	1
q_0	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$
q_1	$\{q_0, q_1, q_2\}$	$\emptyset$
q_2	$\emptyset$	$\emptyset$

Initial State: q0

final states: q0, q1, q2, because the ε-closure each contains q2.

Result: The equivalent NFA has no ε-transitions and accepts the same language as the Orignal of ε-NFA.

Q4. find ε-closure and construct an equivalent DFA

from the given diagram:

- A has a self-loop on input 1
- A has an ε-transitions to B
- A has a transition to B on input 0
- A has a transition to C on input 0
- B has a self-loop on input 1
- B has an ε-transition to C
- C has self-loops on inputs 0 and 1

State	ε-closure
A	{A, B, C}
B	{B, C}
C	{C}

Initial state: A

final State : C

ep 2: construct the DFA

he initial DFA state is {A, B, C}. Applying
he subset construction gives

DFA state	0	1
{A, B, C}	{B, C}	{A, B, C}
{B, C}	{C}	{B, C}
{C}	{C}	{C}

All three DFA states are final because each contains C, the final state of the original ε-NFA

final result

the equivalent DFA contains three reachable states:

{A, B, C} {B, C} {C}

It accepts the same languages as given ε-NFA